



Habib University

ID: Solution

EE-437 Renewable energy systems – Fall 2025

Time = 20 minutes

Final Exam Part 1

Max Points: 20

These are BCQs, meaning it is not necessary to have an exact correct answer. Choose the best answer.

Mark the answer on the question paper. Each question carries equal marks. In case two answers are marked, no answer will be considered.

Q1) Which thermodynamic process is used to define coal power plants?

A) Carnot	B) Brayton	C) Rankine	D) Joule
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Q2) Which thermodynamic process is used to define gas power plants?

A) Carnot	B) Brayton	C) Rankine	D) Joule
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Q3) If a country is located near the north pole, what can be said about solar PV production there compared to a country near the equator?

A) Higher	B) Lower	C) Same	D) Higher or lower
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Q4) Which is the most optimum orientation for a solar panel in Islamabad?

A) North	B) South	C) East	D) West
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Q5) What does the optimum tilt angle of a solar panel depend most on?

A) Zenith	B) Azimuth	C) Latitude	D) Solar altitude
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Q6) If a country is located near the north pole, what can be said about wind production there compared to a country near the equator?

A) Higher	B) Lower	C) Same	D) Higher or lower
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Q7) How does Power of a wind turbine scale with velocity (v) of wind?

A) $P \propto V$	B) $P \propto V^2$	C) $P \propto V^3$	D) $P \propto \ln(V)$
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Q8) What is the maximum power that can be taken out of wind?

A) 33%	B) 66%	C) 95%	D) 100%
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Q9) Is Hydro power the most stable form of renewable energy in the world?

A) Yes	B) No	C) Depends on longitude	D)
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Q10) What factors determine the energy in hydro power plants?

A) Height	B) Flow rate	C) Plant efficiency	D) All of them
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Q11) Why can't water from well be directly used in turbine?

A) Impurities	B) Too high pressure	C) Too high water temp	D) All of them
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Q12) Which of these has the highest deaths per energy produced?

A) Coal	B) Wind	C) Nuclear	D) Gas
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Q13) What impact does biogas collection have on GHG emissions?

A) Positive	B) Negative	C) Highly negative	D) Neutral
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Q14) Which of these has highest self discharge?

A) Flywheel	B) Pumped hydro	C) NiMH	D) Lithium ion
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Q15) Which of these has lowest self discharge?

A) Flywheel	B) Pumped hydro	C) NiMH	D) Lithium ion
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Q16) If a certain lithium cell is 10Wh, what would you expect from a DLC of same size?

A) 8Wh	B) 80Wh	C) 200Wh	D) 0.8Wh
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Q17) Where is lithium located in a lithium-ion cell?

A) Cathode	B) Anode	C) Electrolyte	D) All of them
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Q18) What is the nominal voltage of a lithium-ion cell

A) 3V	B) 3.5V	C) 4.2V	D) 4.7V
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Q19) What happens when 2 cells are connected in parallel?

A) Double voltage	B) Double SoC	C) Double capacity	D) All of them
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Q20) Which power plant has highest load gradient?

A) Coal	B) Gas	C) Nuclear	D) CCGT
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